Traffic Accident Analysis & Exposure-Adjusted Risk: 12-Minute Talking Points

Visual Analytics Lab, Review 1 | Saranya Dalal [26MML0007], Divya Pujari [26MML0009], Rohan Yadav [26MML0020] | Guide: Dr Manikandan K (SCOPE)

Red lines are suggested wording. Green lines are live-demo steps in the Streamlit dashboard. For three speakers, one split is 0:00-4:30, 4:30-8:15 and 8:15-12:00.

0:00 - 1:00 Opening: why crash counts mislead

- Agencies usually rank roads by **how many crashes** happen on them and fund the top of that list.
- Counts mostly track traffic volume. A motorway carrying 50,000 vehicles a day will always log many crashes, most of them low-speed shunts.
- A quiet rural curve may log only a handful, but those crashes are far more likely to kill, and it never reaches the top of a count list.
- So the question is not 'where do most crashes happen' but 'where is each vehicle-kilometre most dangerous'.

Say: “One-eighth the crashes, twelve times the risk. Low frequency masks high severity.”

1:00 - 2:00 Research gap and research questions

- Literature (Slide 4): XGBoost, SVM and DeepSHAP models predict severity well, but most hotspot maps still rank by raw frequency.
- Few studies combine **exposure** (traffic volume x road length) with **severity weighting** in one interactive tool.
- **RQ1 - When?** Do crash volume and crash lethality peak at the same times?
- **RQ2 - Where?** Which segments are true hotspots once exposure is accounted for?
- **RQ3 - Under what conditions?** How do lighting, surface, weather, speed and road-user type change severity?
- Plus a transferability check on Indian mixed traffic (Bengaluru).

2:00 - 3:15 Data and pipeline

- 22,000 crash records in the UK STATS19 schema on 160 road segments (OSM-style attributes: class, lanes, speed limit, lighting, curvature), joined to AADT volumes.
- Why the UK (Slide 7): it is the only context where crash records and official AADT are both open and joinable.
- Pipeline: **01** ingest to Parquet -> **02** preprocessing and features (circadian hour, weekend, night, wet, unlit, vulnerable user) -> **03** exposure and risk in DuckDB -> **04** pre-aggregated cubes (Parquet + GeoJSON) and the ML model.
- The whole batch runs in about 5 seconds. Dashboards read small cubes instead of querying raw rows, so filtering is instant.
- Why DuckDB: embedded and columnar, no server, reads Parquet directly. No Airflow: four sequential scripts are easier to reproduce and defend.

Demo: Pipeline Architecture tab (show the 6-stage diagram).

3:15 - 4:30 Method: exposure-adjusted severity risk

- **Exposure** = AADT x length (km) x 365 / 1,000,000, in million vehicle-km (MVKM). The network totals 8,611 MVKM.
- **Severity weights** from UK DfT injury-cost ratios: Fatal 50, Serious 10, Slight 1. 837 fatal and 4,609 serious crashes in total.
- **Risk** = weighted crash sum / exposure. Network mean is 20.2 per MVKM (median 16.5).
- **Suppression**: segments with <3 crashes or <0.2 MVKM are flagged, because tiny denominators make risk swing wildly.
- **Four quadrants** from median splits on crash count and risk: Q1 severe congested, Q2 deadly trap, Q3 safe, Q4 fender-benders.
- **Rank shift** = count rank minus risk rank. A large positive shift means raw counts hide the danger.

4:30 - 6:00 Core finding: frequency and severity point to different roads

- **SEG-0417, M25 interchange**: 47 crashes but only 6.86 per MVKM, risk rank #152. Heavy traffic dilutes the risk.
- **SEG-1902, B4009 Chiltern ridge curve**: 6 crashes, 84.4 per MVKM, risk rank #2. That is 0.13x the crashes and 12.3x the risk.
- By road class: motorways carry 31% of all crashes (6,857) but have the lowest mean risk (9.8). Rural B-roads have 1% of crashes (220) and the highest mean risk (29.4).
- Of the 10 busiest segments by count, 9 are Q4 fender-benders with a median risk rank of 142. Of the 10 riskiest, 9 are Q2 deadly traps with a median count rank of 117.
- 48 of 160 segments are deadly traps. A count-based budget would spend first on the roads that are safest per kilometre driven.

Demo: Core tab. Point at the two benchmark cards, then the top-left of the quadrant scatter.

Say: “The two rankings are nearly mirror images. That is the core contribution.”

6:00 - 7:15 RQ1 - When do crashes happen vs. when are they lethal?

- Volume follows commuting: 08:00 is the busiest hour (1,908 crashes) and 17:00 is next (1,829).
- Lethality runs the other way: 05:00 has the highest fatal share (6.8%) and mid-afternoon the lowest (15:00, 2.6%).
- Night (20:00-05:59) fatal share is 5.3% against 3.5% by day: emptier roads, higher speeds, darkness and fatigue.
- Friday and Saturday late nights (22:00-03:59): 28.5% of crashes are fatal or serious against 24.7% at other times.
- The 168-cell day x hour heatmap shows the hot cells sit in the small hours, e.g. Sunday 01:00 (46% severe).

Demo: WHEN tab. Hover the dual-axis chart, then switch the heatmap metric between count and severe rate.

Say: "Peak traffic is not peak danger."

7:15 - 8:15 RQ2 - Where are the true hotspots?

- 3D map of all segments. The colour toggle switches between raw count, exposure-adjusted risk and rank shift; column height is risk.
- Biggest upward shifts are rural B-roads: SEG-1902 +157, SEG-0142 +150 (count rank #151 to risk rank #1), SEG-0052 +134.
- Biggest downward shifts are motorways: SEG-0046 is #1 by count (462 crashes) but #158 by risk.
- Road-class and suppression filters in the sidebar update the map and the leaderboard together.

Demo: WHERE tab. Switch 'Color segments by' to Rank Inversion Shift and hover one red segment.

8:15 - 9:15 RQ3 - Under what conditions?

- **Lighting matters most**: in unlit darkness 37.3% of crashes are fatal or serious, against 24.1% in daylight and lit darkness.
- Unlit segments average 23.1 risk per MVKM against 18.5 for lit ones. Severe blind bends average 37.0.
- **Vulnerable road users**: motorcycle crashes are 44.8% severe against about 21% for cars; at 70 mph it is 54% over 1,689 crashes.
- **Weather and surface matter less than expected**: fine, rain, fog and snow are all 23-25% severe; dry and wet roads both about 25%.

Demo: UNDER WHAT CONDITIONS tab (surface x lighting chart, then speed-limit chart).

Say: "Street lighting and speed are the levers, more than the weather."

9:15 - 10:15 ML classifier and what-if simulator

- Random Forest trained on all 22,000 records; target is fatal or serious (24.75% of crashes).
- Top drivers: vulnerable road user (0.37), AADT (0.15), speed limit (0.12), segment length (0.11), dark and unlit (0.07).
- The simulator is a policy sandbox. It uses additive weights chosen to mirror these drivers, not live model predictions.
- Walk-through: 60 mph + vulnerable user = 50% (moderate). Add dark night and no street lighting: 90% (high), and the advice changes to LED lighting, speed cameras and high-friction surfacing.

Demo: ML tab. Change lighting to Unlit and time to Dark Night live.

10:15 - 11:15 India transferability case (Bengaluru)

- The **Study Context** switch in the sidebar re-points every tab to 5 Bengaluru corridors with 2,500 crashes.
- Mixed traffic: two-wheelers are 45-68% of corridor volume, and two-wheelers plus pedestrians are 58% of crashes.
- Night fatal share is 7.1% against 4.9% by day, a wider gap than in the UK.
- Same inversion: Sarjapur Road has 0.97x the crashes of Outer Ring Road but 4.8x the risk (49.96 vs 10.42 per MVKM).
- Kanakapura NH-948 (unlit peripheral curves) has the lowest AADT (18,000), the most fatalities (55) and the highest risk (56.8).
- Data gaps: two-wheelers are exempt from FASTag tolls, so volumes need expansion factors; the feed has no weather or lighting fields.

Demo: switch Study Context to India, show the Core and WHEN tabs, then the India tab.

Say: "The method transfers: low frequency still masks high severity in Indian traffic."

11:15 - 11:45 Power BI version, limitations and next steps

- The same cubes also feed a 7-page Power BI report with synced study-context, road-class and suppression slicers.
- Limitations: records are generated in the STATS19 schema; AADT is a single annual figure; quadrants use median splits; the simulator is heuristic.
- Next steps: load real STATS19 and DfT AADT, add empirical-Bayes smoothing for low counts, SHAP explanations, and real Bengaluru police data.

11:45 - 12:00 Close

- Counts tell you where crashes happen. Exposure-adjusted severity tells you where people die per kilometre driven.

Say: "Rank roads by risk per vehicle-km, not by crash count, or the deadliest roads stay hidden. Thank you."

Likely questions

If asked	Short answer
Is the data real?	Pipeline stage 1 generates the records in the STATS19 schema with fixed seeds. A real STATS19 extract can replace them without code changes.
Why 50 / 10 / 1?	They follow UK DfT injury-cost ratios. Changing the weights changes the scale of risk, not the main inversion.

Why median splits?	They are simple and explainable for Review 1. Percentile or risk-threshold cut-offs are a planned extension.
Why suppress segments?	Very small counts or exposure make risk swing wildly, so those segments are flagged instead of ranked.
Is the simulator the ML model?	No. It uses transparent additive weights that mirror the model's top drivers, so engineers can see why the score moves.
Why DuckDB, not Postgres?	Embedded, columnar, no server, reads Parquet directly, and runs the window functions in under a second.
Does Q2 mean each crash is deadlier?	Not necessarily. Q2 means more severity-weighted harm per vehicle-km, which is what matters for prioritising money.

Demo checklist

- Start **streamlit run app.py** before the session and keep the browser tab open (the cubes are already built).
- Start on the Core tab with Study Context = UK, all road classes selected, suppression on.
- Switch back to UK after the India demo so later questions start from the main dataset.
- Backup if the live demo fails: report_figures/ (fig04 quadrants, fig07 circadian, fig10 map, fig19 simulator, fig20 Bengaluru).